

Selective Inference for Weights-Matrix Selection

Choosing the Neighbour Parameter by Inspecting the Result Inflates the
False-Positive Rate, and the Correction Is Derivable

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Abstract

[Placeholder abstract — T-009.] A permutation test for spatial association builds its null distribution by reshuffling the observed values across locations. That construction assumes the neighbour structure was fixed before the data were seen. In practice the neighbour parameter is routinely chosen by trying several values and keeping the one that gives the strongest signal. This paper shows that the practice inflates the false-positive rate by roughly a factor of three, measures the inflation as a surface over selection rules and candidate grids rather than as a point, and derives the correction. Numbers, figures and citations in this draft are placeholders until the evidence tasks land; every one of them will be traceable through `papers/CLAIMS.md`.

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1 How to read this paper

This paper is written as lecture notes that happen to contain a new result. It assumes a numerate reader — algebra and reading a graph — and no statistics background at all. Every term of art is defined before it is used, and collected in the glossary in Section 11.

Because the exposition is long by design, each section is tagged with its role, and the tag appears in the table of contents and in the PDF bookmarks. A reader who already knows the background can jump straight to the sections tagged [CONTRIBUTION].

Section	Role	What it is for
2	exposition	Why choosing k by looking at the answer is a problem.
3	exposition	Everything statistical the argument needs, from scratch.
4	exposition	The failure, stated as a concrete procedure.
5	contribution	Why selection breaks exchangeability.
6	evidence	The inflation surface over rules and grids.
7	contribution	The corrected test and its derivation.
8	contribution	What is new here, against prior work.
9	limitations	What the claim does not cover.

Table 1: Role of each section. Sections tagged *contribution* carry the new result; the rest can be skipped by a reader who already has the background.

2 Introduction [EXPOSITION]

[Placeholder — T-009.] The permutation test is the workhorse of applied spatial statistics precisely because it makes so few assumptions. This paper is about the one assumption it does make and that practitioners routinely break without noticing. The statistic under test is Moran’s I (Moran, 1950); the local framework is due to Anselin (1995); all sources are annotated in Section 10.

Figure 1 shows the shape of the effect.

3 A primer, from first principles [EXPOSITION]

This section is shared verbatim with the companion paper on component structure, so that each paper can be read on its own. Readers who have read the other paper can skip it.

3.1 What this primer assumes

This primer assumes a reader who is comfortable with algebra and with reading a graph, and who has no background in statistics whatsoever. Every term of art is defined before it is used. Terms are set in ***bold italic*** at the point of definition and collected in the glossary at the end of the paper, each with the section number where its definition lives.

3.2 Locations, values, and neighbours

Suppose we have measured one number at each of n places — a rate, a count, a score. Write y_i for the value at place i . Spatial statistics asks whether the arrangement of those numbers across the map carries information that the numbers alone do not.

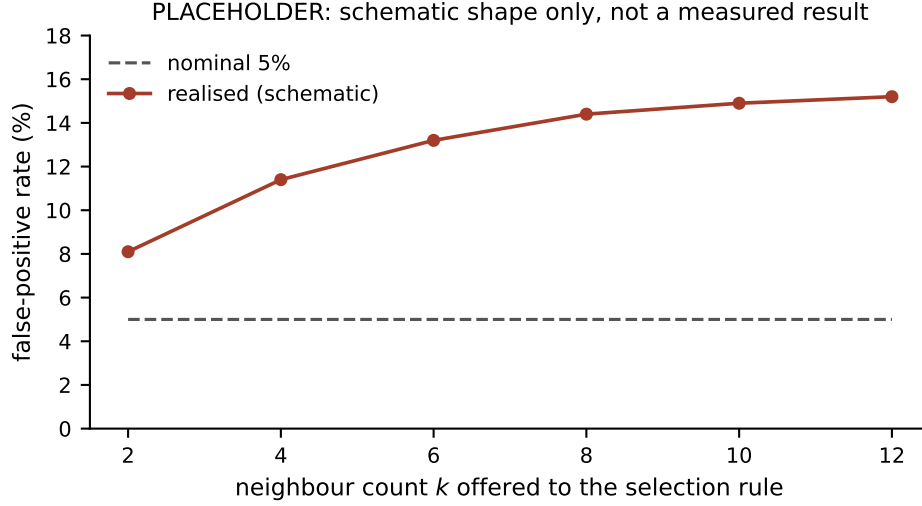


Figure 1: **[Placeholder figure.]** Schematic of the realised false-positive rate against the nominal five per cent, as a function of how many candidate k values the selection rule is offered. Generated by `papers/shared/mkfigs.py`; replaced by the real measured surface in the evidence tasks.

To ask that question we must first say which places count as neighbours of which others. That bookkeeping is done by a *spatial weights matrix*: a square table $\mathbf{W}$ with one row and one column per place, whose entry w_{ij} records how strongly place j counts as a neighbour of place i . A common choice sets $w_{ij} = 1$ when j is one of the k nearest places to i and $w_{ij} = 0$ otherwise.

Reading the table as a map of who-is-connected-to-whom, a group of places that can all reach one another by hopping along nonzero weights, and which no weight connects to anything outside the group, is a *connected component* of the neighbour graph. A neighbour graph with more than one component has been cut into pieces that never exchange information.

3.3 Evidence against a hypothesis

Statistical tests work by contradiction. We name a dull explanation for what we observed, then ask how surprising the observation would be if that dull explanation were true. The dull explanation is the *null hypothesis*; here it is usually “the values were scattered across the map with no regard to location.”

The measure of surprise is the *p-value*: the probability of seeing a result at least as extreme as the one observed, computed under the assumption that the null hypothesis holds. A small p-value means the observation sits far out in the tail of what the dull explanation predicts. It does not mean the dull explanation is false, and it is not the probability that the dull explanation is false.

Placeholder. This is placeholder primer content supplied by the build scaffold (T-005) so that the document compiles, the glossary resolves, and the cross-reference machinery can be checked. T-004 replaces this file with the real from-first-principles exposition required by AC-003.

4 The problem [EXPOSITION]

[Placeholder — T-009.] The failure stated as a procedure a practitioner would actually run: build the weights matrix at several values of k , compute the statistic at each, report the best one and its p-value.

5 The mechanism [CONTRIBUTION]

[Placeholder — T-009.] A permutation test builds its null by reshuffling observed values across locations, so it assumes exchangeability — and exchangeability quietly requires that the weights matrix was fixed in advance. Selecting k from the data conditions the reported statistic on the event that it was the maximum, and the unconditional null no longer describes it.

Key insight. The reported p-value answers the question “how unusual is this statistic?” when the question actually asked was “how unusual is the largest of several statistics?” The candidate statistics are strongly correlated, so the inflation is smaller than a naive multiple-comparisons bound but far from negligible.

6 Evidence [EVIDENCE]

[Placeholder — T-003 supplies the numbers, T-009 the prose.] Type I error inflation across at least three selection rules — arg-max of I , minimum p-value, first significant — and at least two candidate grids, reported as a surface rather than a point (AC-007), at a replicate count sized for a Monte Carlo standard error below half a percentage point (A-4).

7 The correction [CONTRIBUTION]

[Placeholder — T-009.] The corrected test, derived rather than asserted, together with what it costs in power and when the naive test is safe to keep using.

8 What is new here [CONTRIBUTION]

[Placeholder — T-009, informed by the prior-art check in T-001.] This section states the novel contribution and distinguishes it from the general post-selection inference machinery it specialises (AC-008). Where the literature check finds work that partially preempts the claim, the claim is narrowed here rather than the source omitted (D-008).

What is new. Placeholder. The claim stated here must survive the prior-art check, and must be narrowed if it does not.

9 Limitations [LIMITATIONS]

[Placeholder — T-009.] Which claims rest on a single dataset, which are simulation-based, and what observation would falsify them (AC-010). In particular, the inflation figure is measured under a specific null and a specific family of candidate grids, and does not transfer automatically to other selection procedures.

10 Annotated bibliography

The reference list that follows these annotations is generated by BIBTEX from the shared file `shared/refs.bib`. Each entry below states what the source does, why it is credible, and precisely what the present paper takes from it.

Moran (1950) *What it does:* introduces the autocorrelation coefficient now universally called Moran's I , as a way of detecting whether a quantity measured over a region varies smoothly or erratically in space. *Why it is credible:* the founding paper for the statistic, in *Biometrika*, and the definition in use today is still Moran's. *What we take:* the definition of the statistic and its ratio form.

Getis and Ord (1992) *What it does:* defines the family of distance statistics G_i and G_i^* for detecting local concentrations of high or low values. *Why it is credible:* the originating paper in *Geographical Analysis*; the statistic is implemented under this definition in every major spatial toolkit. *What we take:* the definition of G_i^* , including the fact that it standardises against the *global* mean and standard deviation.

Anselin (1995) *What it does:* sets out the LISA framework, which decomposes a global measure of spatial association into per-location contributions. *Why it is credible:* the standard reference for local spatial statistics, and the basis of the local Moran implementations in `spdep`, `esda` and ArcGIS. *What we take:* the framing of local statistics as decompositions of a global one.

Placeholder. This is a stub annotated bibliography supplied by the build scaffold (T-005) to prove the BIBTEX wiring. T-001 replaces this file, and `refs.bib`, with the retrieved and verified corpus required by AC-004 and AC-005.

References

- Luc Anselin. Local indicators of spatial association—LISA. *Geographical Analysis*, 27(2): 93–115, 1995. doi: 10.1111/j.1538-4632.1995.tb00338.x.
- Arthur Getis and J. K. Ord. The analysis of spatial association by use of distance statistics. *Geographical Analysis*, 24(3):189–206, 1992. doi: 10.1111/j.1538-4632.1992.tb00261.x.
- P. A. P. Moran. Notes on continuous stochastic phenomena. *Biometrika*, 37(1/2):17–23, 1950. doi: 10.2307/2332142.

11 Glossary

Every term of art used in this paper, with the section in which it is defined.

Connected component A group of places in the neighbour graph that can all reach one another by hopping from neighbour to neighbour, and which is joined to nothing outside the group. A graph with two components has been cut in two. (*Defined in Section 3.2, page 3.*)

Null hypothesis The deliberately dull explanation a statistical test tries to contradict — here, that the observed values were scattered across the map with no regard to location. (*Defined in Section 3.3, page 3.*)

p-value The probability of seeing a result at least as extreme as the one observed, computed on the assumption that the null hypothesis is true. A small p-value says the observation is unusual under the dull explanation; it is not the probability that the dull explanation is wrong. (*Defined in Section 3.3, page 3.*)

Spatial weights matrix A square table with one row and one column per place, whose entry in row i and column j records how strongly place j counts as a neighbour of place i . It is the object that turns a map into a graph. (*Defined in Section 3.2, page 3.*)